

Automatic exercise detection and basal insulin adjustment for type 1 diabetes

A machine learning proposal. Two-page summary.

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The problem

I have had type 1 diabetes for over twenty years. My insulin pump and glucose sensor work together as a hybrid closed-loop system, and at rest they keep my glucose in range without any input from me. Exercise breaks that. I get low glucose alarms minutes into a gym session, and after an evening of football or a day of cricket I often go low overnight.

This is common. Over 60% of people with type 1 diabetes do not reach recommended activity levels, and fear of low glucose is a main reason (Cockcroft et al., 2020). On exercise days time in range improves from 70% to 76%, while time below range nearly triples (Riddell et al., 2023).

Where the current system fails

1. **It relies on the user.** The pump's activity setting has to be switched on by hand, and that happens before only around 20% of sessions (Turner et al., 2025).
2. **It treats everyone the same.** The setting applies one fixed raised target, yet the same person doing the same exercise gets a different glucose response almost every session (Riddell et al., 2023).
3. **It reacts late.** The algorithm works from sensor readings that lag behind blood glucose, so by the time it responds the drop is already under way.
4. **It forgets the exercise afterwards.** Insulin sensitivity stays raised for 24 to 36 hours, which is when the overnight lows happen.

The proposed system

The proposal joins three pieces of published research into one system.

- **Detect exercise automatically** from heart rate and movement data, which consumer watches already record. Jacobs et al. (2015) identified exercise this way with 97% sensitivity.
- **Predict where glucose is heading** with an LSTM, a type of neural network suited to sequences over time. Shao et al. (2024) predicted mild low glucose from sensor data with an AUC of 97%.
- **Adapt to the individual** with transfer learning. A model trained on many patients is fine-tuned on one person's data (Thomsen et al., 2025).

Each piece has only been tested on its own, and joining them would be a significant engineering job. The patient can override the system at every stage.

How it would be trained and introduced

The dataset this needs does not exist yet. I contacted Tidepool, whose data donation project was the obvious source, and they confirmed it holds no heart rate or movement data. So the proposal starts with a companion app that collects watch data from consenting participants and links it to their glucose and insulin records. Training then runs in three phases.

Phase	What the model does	What the patient does
1. Population training	Learns general exercise and glucose patterns from several hundred recruited patients	Uploads data
2. Shadow mode	Detects exercise, predicts glucose and suggests an insulin adjustment, but does not act. Tunes itself to the individual.	Sees each prediction with a confidence level and compares it with what happened
3. Active	Adjusts background insulin itself. Falls back to sensor data alone if the watch drops out.	Can override at any point

Shadow mode does two jobs. It tunes the model to the person, and it lets the person learn when to trust the model and when to overrule it before the model has any control. Under the EU AI Act a system like this is high-risk, so each phase is designed around human oversight, data governance and conformity assessment.

Risks and bias

- **Who is in the data.** People who donate data tend to be engaged with technology and able to afford sensors and watches. Older, less active and less well-off patients would be under-represented, and they may have the most to gain. Recruiting through clinics narrows the gap without closing it.
- **A model that cannot explain itself.** The reasoning inside an LSTM cannot be fully inspected. The proposal shows a confidence level with every prediction and flags low-confidence adjustments for clinical review.
- **Over-trust.** Even at 95% sensitivity, 5 in every 100 severe lows go undetected (Shao et al., 2024). If the patient has learned to defer to the system, nothing catches them. The threshold has to be set to catch nearly all severe events and accept more false alarms, while keeping alarm fatigue in check.

What I learned

I expected the algorithm to be the hard part. The real barriers were data that does not exist, recruitment and regulation. That moved me from thinking about the algorithm first to thinking about the whole system first, including the people who have to trust it and the rules it has to meet.

The study in this repository is a first practical step. It uses my own sensor, pump and watch data. It is analysis only and is never used for dosing.

Selected sources

Cockcroft, E.J., Narendran, P. and Andrews, R.C. (2020) Exercise-induced hypoglycaemia in type 1 diabetes. *Experimental Physiology*, 105(4).

Jacobs, P.G. et al. (2015) Incorporating an exercise detection, grading, and hormone dosing algorithm into the artificial pancreas. *Journal of Diabetes Science and Technology*, 9(6).

Riddell, M.C. et al. (2023) Examining the acute glycemic effects of different types of structured exercise sessions in type 1 diabetes (T1DEXI). *Diabetes Care*, 46(4).

Shao, J. et al. (2024) Generalization of a deep learning model for continuous glucose monitoring-based hypoglycemia prediction. *JMIR Medical Informatics*, 12.

Thomsen, H.B. et al. (2025) Prediction of hypoglycemia from continuous glucose monitoring using transfer learning. *Journal of Diabetes Science and Technology*, 19(3).

Turner, L.V. et al. (2025) Use of the Omnipod 5 automated insulin delivery system activity feature. *Diabetes Care*, 48(9).

The full proposal, with all references, is available on request.